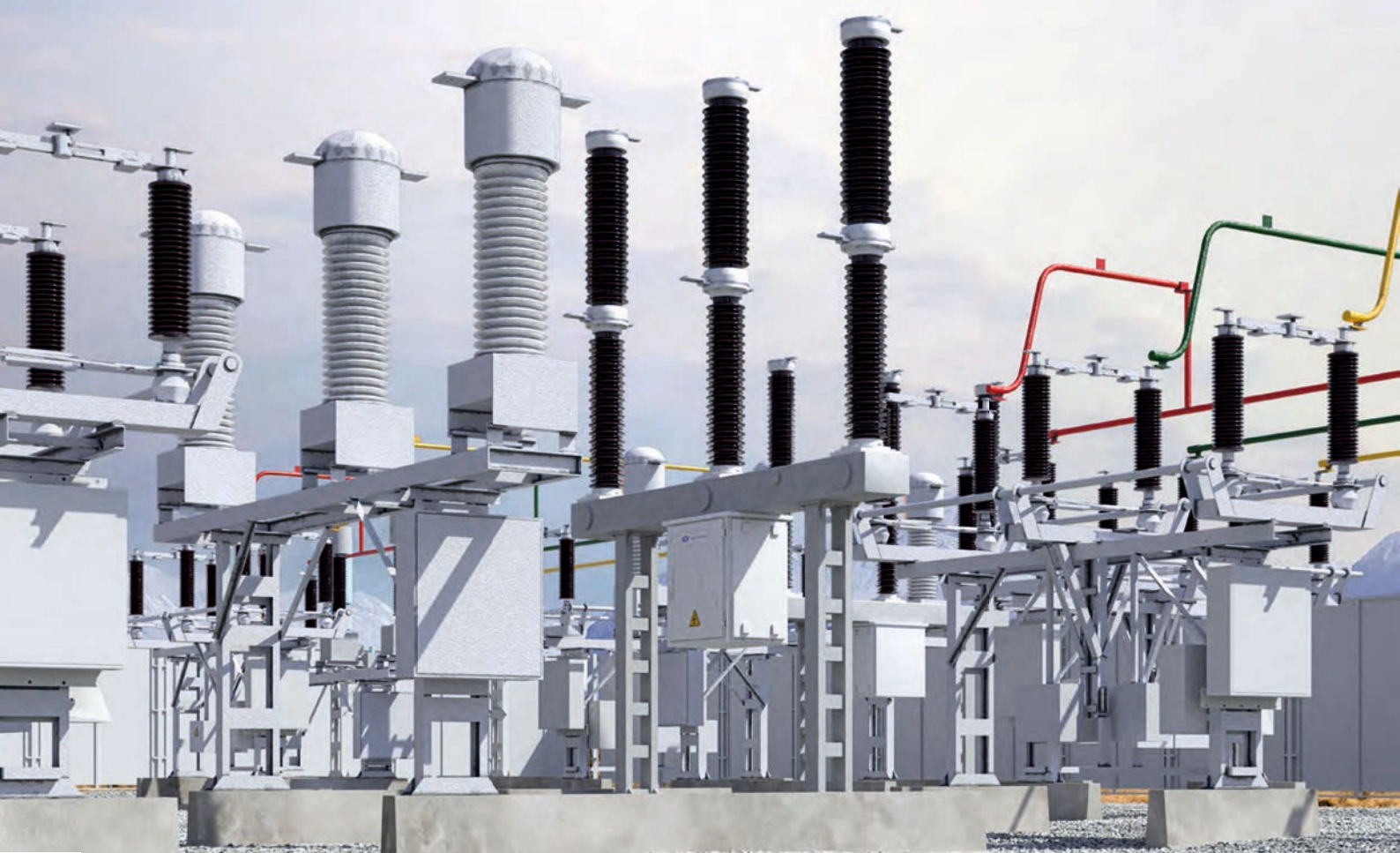


CATALOGUE OF ELECTRICAL EQUIPMENT



UZELEKTROAPPARAT

2022



UZELEKTROAPPARAT

87

YEARS

**OF SOLUTIONS, IDEAS
AND EXPERIENCE**

Our equipment is used in residential buildings, in production, on ships, submarines, aircraft, on oil and gas pipelines, railways, in agriculture and waterside structures, at thermal, hydro and nuclear power plants and electrical networks.

We successfully cooperate with our partners in a number of power supply projects in the energy, mining, oil and gas, transport, chemical, automotive, construction sectors, both in our country and abroad.

In addition to internal quality control, our products comply with national O'zDSt standards and interstate GOST standards, as well as IEC (International Electrotechnical Commission) international standards are used in the manufacture of our products. The radio communication equipment produced by our company is certified by the European standards ETSI (European Telecommunications Standards Institute), DMR (Digital Mobile Radio) and TETRA (TERrestrial Trunked Radio). The company is certified according to international standards ISO 9001; ISO 14001 and OHSAS 18001.

THE GROUP OF COMPANIES COOPERATES WITH MORE THAN 30 COUNTRIES AND THIS IS NOT THE LIMIT FOR US.

Today, our company supplies complex solutions - from design to commissioning of equipment.

In its strategy, the group of companies relies on the continuous improvement of product quality and reliability, paying attention to environment and the rational use of natural resources.

The main difficulty in the construction and launch of any production facility is a thoughtful and verified way of supplying electricity without harm to the environment and reducing the quality of supply to the existing infrastructure.

1941

UZELEKTROAPPARAT
was founded

450 types

of electrical equipment
are produced by a group of companies

5 000

substations produced annually

1 000 000+

meters are manufactured per year

12

shops involved in production

30+

of partners world leading companies

500 kV

max voltage of manufactured equipment

4 500

production staff

8.2 hectares

the territory of the plant

230 000+

kg of copper consumed per quarter

\$341 000 000

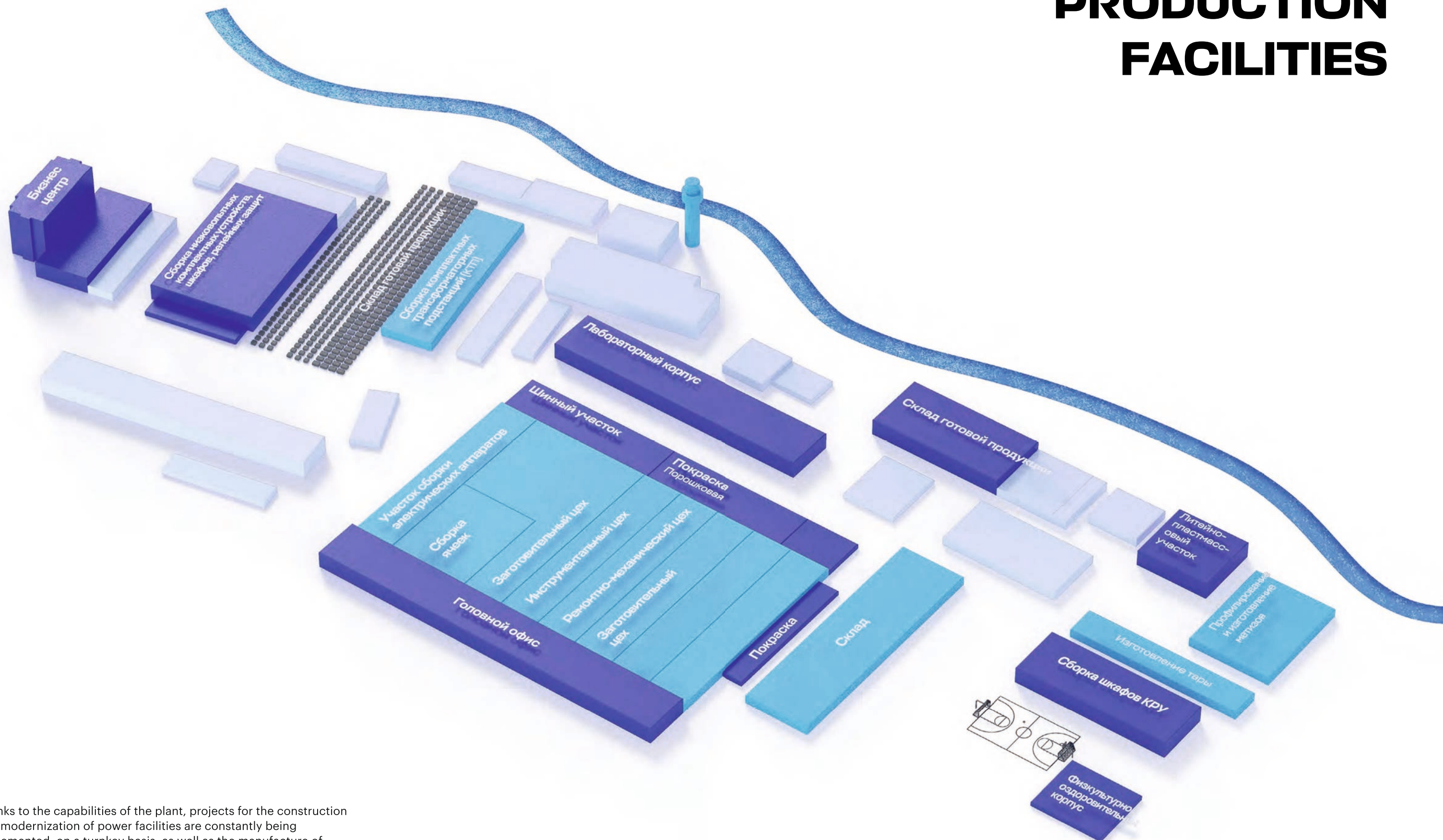
company turnover for 2020

COMPANY HISTORY

1941	◆	Date of foundation of Uzelektroapparat, which at that time was called Electrostanok (evacuated plant from Kharkov). In 1942, the output plan for the first quarter was already fulfilled by 182%, for the second - by 224%, for the third - by 312%, and 331% for the fourth.
1942	◆	Date of foundation of the Chirchick Transformer Plant (ChTZ), at that time the plant was called Electroshield plant, which during the war years was the only enterprise in Central Asia that produced electrical panels, power transformers, substations and other electrical installations.
1946	◆	The Electrostanok plant and the High Frequency Current Plant (evacuated from Moscow) are merged in the Tashkent Electrotechnical Plant (TEZ).
1949	◆	TEZ becomes the main enterprise in the USSR for the production of multi-purpose package switches for currents from 10 to 400 Amperes. In 1950, there were already 2.5 times more package switches, and 6 times more tubular fuses than in 1948.
1952	◆	For the first time in the USSR, 6 kV switchgears (KRU) were developed and manufactured at the Eletroshield plant.
1961	◆	TEZ, in collaboration with the Electroshield plant, produced the first 620 units of electrical panels. Four years later, 9169 electrical panels were already manufactured, that is, almost 15 times more.
1962	◆	The Electroshield plant was renamed into the Chirchik Transformer Plant (ChTZ).
1963	◆	From 1963 to 1965, low-voltage package equipment and an already large range of electrical devices manufactured by TEZ put into operation more than 400 large new constructions of the country: Tashkent and Navoi Power Plants, Sumgait and Nizhnekamsk Thermal Power Plants, Altyn-Topkansky, Sokolovsky-Sarbaysky and Kachkanarsky Mining and Processing Plants, West Siberian Metallurgical Plant and others.
1966	◆	The branch of the State Design and Technology Bureau "Electroprivod", based on the design and technology departments of the TEZ, was transformed into a branch of VNIIR (All-Union Research Institute) "Electroprivod" and becomes the head organization of the Ministry of the Electrical Industry for low-voltage equipment for power plants and substations.
1969	◆	For the development and introduction into production of the RPU-1 control relay and the PKP-U-3-12 apparatus, a group of designers and production workers was awarded the First Degree Diploma of VDNKh.
1971	◆	The Tashkent Electrotechnical Plant was renamed into the Sredazelektroapparat.

1973	◆	The staff of "Sredazelektroapparat" received the First author's certificate for the invention "Contact system of an electromagnetic relay" (Authors V.M. Dvorin, L.N. Gargalas and O.A. Rybnikov).
1979	◆	The VNIIR department was reorganized into the Special Design and Technological Bureau for Low-Voltage Complete Protection Devices and System Automation for the Power Industry (SPKTBE) and became part of the Sredaselektroapparat Production Association as a legal entity.
1991	◆	Mr. Vladimir Royuk becomes the youngest commercial director in the history of the enterprise. Huge energy and an inexhaustible supply of ideas of Mr. Royuk allowed in the difficult time of the early 90s to restore the enterprise, provide it with work and give people the opportunity to believe that the best is yet to come. He kept the traditions and human relations at the enterprise, which were from the date of foundation.
1993	◆	Sredazelektroapparat was renamed Uzelektroapparat.
2000	◆	On the basis of Uzelektroapparat, two joint ventures with foreign investments were formed: JV Tashelektroapparat and JSC Uzelektroshield. Vladimir Royuk was elected General Director of JV Tashelektroapparat.
2007	◆	The Uzelektroapparat was renamed into the Uzbek-Russian Open Joint Stock Company Uzelektroapparat-Electroshield. The international certificate ISO 9001:2008 was issued. On November 30, the First International Scientific and Practical Conference "Design and production of complete devices and electrical equipment for power plants and industrial enterprises." took place.
2008	◆	A large-scale modernization of production technology has been carried out. The company has installed technological lines of companies: Prima Power, FINN POWER, Bystronic, Ermaksan, TAISS, Haas, etc.
2010	◆	For merits in strengthening the country's economic potential and successful implementation of reforms, many years of fruitful work and a significant contribution to the development of the country's economy, active participation in public life, Vladimir Royuk was awarded a government award - the Dustlik Order.
2017	◆	On behalf of the President of the Republic of Uzbekistan, the NUKUSELECTROAPPARAT LLC was founded.
2021	◆	According to the decree of the President of the Republic of Uzbekistan, the General Director of JSC UZELEKTROAPPARAT-ELECTROSHIELD, Vladimir Royuk was awarded the honorary title "Honored Worker of Industry of the Republic of Uzbekistan".

PRODUCTION FACILITIES



Thanks to the capabilities of the plant, projects for the construction and modernization of power facilities are constantly being implemented, on a turnkey basis, as well as the manufacture of electrical equipment for stations and substations, and the design of power facilities. In addition, the group of companies has a license to design engineering networks, systems of buildings and structures, high-risk facilities, electrical equipment for stations

WORLDWIDE OPPORTUNITIES

OVER 80 YEARS OF WORK AND DOZENS OF MAJOR PROJECTS IMPLEMENTED, THE GROUP OF COMPANIES HAS EXPANDED ITS GEOGRAPHY OF SUPPLY FAR OUTSIDE THE REPUBLIC OF UZBEKISTAN. THE PARTNERS OF UZELEKTROAPPARAT ARE THE LEADING FOREIGN COMPANIES IN CHINA AND GERMANY, FRANCE AND RUSSIA, SLOVENIA, THE NETHERLANDS AND OTHER COUNTRIES.

Our equipment is used in residential buildings, in production, on ships, submarines, aircraft, on oil and gas pipelines, railways, in agriculture and waterside structures, at thermal, hydro and nuclear power plants and electrical networks. The company has modern metalworking machines Bystronic (Switzerland), Finn Power (Finland), Haas processing complex (USA), Tristan powder coating lines (Russian Federation) and Taiss (Italy), zinc-based galvanic coating line Galvour (Sweden) and others.



HIGH VOLTAGE EQUIPMENT AND SUBSTATIONS WITH A VOLTAGE OF 10-500 KV

INCLUDING:

Packaged modular transformer substations, with a voltage of 20-500 kV;
Packaged portable transformer substations, with a voltage of 20-35 kV, with a capacity of up to 1600 kVA;
Packaged universal transformer substations, with a voltage of 20-35 kV, with a capacity of up to 630 kVA.

High-voltage equipment and substations are designed to receive, convert and distribute electrical energy of three-phase alternating current with a frequency of 50 Hz and are used to supply industrial and municipal consumers, agricultural areas and large construction sites, as well as on the 500, 220, 110 kV side of large network substations and electrical stations.



Packaged transformer substations 6-(10) kV,
with a capacity of up to 2500 kVA;

INCLUDING:

- 6 (10) kV mast and pole type packaged transformer substations, with a capacity of up to 250 kVA;
- 6(10) kV rural type packaged transformer substations, with a capacity of up to 250 kVA
- 6 (10) kV kiosk type packaged transformer substations, with a capacity of up to 1000 kVA
- 6 (10) kV urban type packaged transformer substations, with a capacity of up to 1000 kVA
- 6 (10) kV industrial type packaged transformer substations, with a capacity of up to 2500 kVA

Packaged transformer substations are designed to receive, convert and distribute electrical energy of three-phase alternating current with a frequency of 50 Hz and are used to supply industrial and municipal consumers, agricultural areas and large construction sites.



Key specifications

Capacity	HV voltage	LV voltage
up to 2500 kVA	6, 10 kV	0.4 kV

K-149 SERIES
HIGH VOLTAGE PACKAGED DEVICES

Switchgear cabinets are designed to operate as part of switchgear in three-phase alternating current networks with a frequency of 50 Hz, and rated voltage of 6 or 10 kV with an isolated and partially grounded neutral through an arcing reactor or resistor.

Including:

- 6 (10) kV switchgears K-61 series for indoor installation, for currents up to 4000 A;
- 6 (10) kV switchgears K-63 series for indoor installation, for currents up to 1600 A;
- 6 (10) kV switchgears K-59 series for indoor and outdoor installation, for currents up to 3150 A;
- 6 (10) kV switchgears K-129 series for indoor installation, for currents up to 1600 A;
- 6 (10) kV switchgear K-XXVI series for indoor installation, for currents up to 2500 A;
- 6 (10) kV prefabricated chambers KSO-2009 series, with one-side maintenance, for currents up to 1000 A (1600 A);
- 6 (10) kV prefabricated chambers KSO-285M, KSO-298 series, with one-side maintenance for currents up to 1000 A;
- 6 (10) kV prefabricated chambers KSO-366, KSO-386 series, with one-side maintenance for currents up to 1000 A;
- 6 (10) kV switchgears KRN series for outdoor installation, for currents up to 1000 A.



Key specifications

Rated voltage	Rated current	Short circuit current
6, 10 kV	up to 4600 A	up to 31.5 kA

K-65M, K-65A, K-65EVA switchgears

The switchgear is designed to receive and distribute electrical energy of three-phase alternating current of industrial frequency 50 Hz for a voltage of 35 kV.



Key specifications

Rated voltage

35 kV

Rated current

up to 4000 A

Short circuit current

**up to 40 kA
with SF6 modules**

Design

**indoor
and module type**

Voltage 6 kV, Capacity – up to 1600 kVA

PACKAGED EXPLOSION-PROOF TRANSFORMER SUBSTATIONS

KTPV Series

An explosion-proof transformer substation is a type of substation, with a capacity from 100 to 1,600 kV A, designed to supply three-phase alternating current in underground mines, where work can be complicated by dangerous proximity to gas and dust.

This type of transformer substations is intended for operation at temperatures from -10 to +35°C and in an environment without shocks and shocks.

Key specifications

Rated capacity

1600 kVA

Rated primary voltage

6 kV

Windings connection group

Y/Y-0; Y-Δ-11

Protection degree

PB Ex db [ia] IX



Voltage 6 kV

Explosion-proof switchgears



KRUV series

KRUV-6V Cabinets are designed for power distribution of common industrial frequency, as well as for protection of electrical equipment with isolated neutral. The standard operating conditions for cabinets are altitudes not exceeding 1000 m above sea level and temperatures ranging from -10 to +35°C. The cabinets provide prompt closing and opening operations in automatic and manual modes, being a set of explosion-proof devices.

As well as:

- 6 kV mine switchgears KRURN series;
- Explosion-proof feeder switches AVF series;
- Mine starters PRN series;
- Mine starting Units APSH series;
- Explosion-proof lamps for mining works;
- 6 kV switchgears for excavators 2KVE and KRUE series;
- LV switchgears for excavators KERU series;
- 6 kV stand-alone cubicle quarry switchgear YAKNO, YAVKO series;
- Mine ventilation systems.

Key specifications

Rated voltage

6 kV

Protection degree

PB Ex d [ia] I

6(10); 0.4 kV soft starters

Electric drive equipment



The soft starter of the UPP series is designed for shockless start of asynchronous and synchronous electric motors, mechanisms with a "fan" characteristic of the load moment in centrifugal compressors, pumps, fans, exhausters and other similar mechanisms.

The soft starter has a digital control system that provides convenient programming of parameters and is made on the principle of a thyristor voltage regulator that provides voltage rise rate limitation.

Key specifications

Voltage	Rated current of electric motor	Power circuits Max. current
0.4; 6; 10 kV	110; 200; 250; 300; 360; 400; 450; 500; 600 A	440; 800; 1000; 1200; 1440; 1600; 1800; 2000; 2400 A

Equipment for frequency regulation

UChR series



As well as:

- 6(10) kV Frequency controller UCHR-6(10) series
- Thyristor exciter VTC series for units with excitation current up to 10,000 A;
- Control cabinets for submersible pumps «Cascade» series

Frequency controller converts a three-phase alternating voltage with a constant frequency and amplitude into a voltage with a variable frequency.

Frequency controller is designed for stepless speed control of asynchronous motors, including electric motors of compressors, pumps, fans with a rated voltage of 380 V.

The use of frequency controller can significantly reduce energy, repair and operating costs while maintaining the same performance of machines, electric motors and mechanisms by optimizing its operation in a wide range of load changes.

Key specifications

Rated input current	Rated output current	Electric motor
3,5 — 650 A	3,5 — 650 A	1,1 - 355 kW

CABINETS FOR RELAY PROTECTION AND AUTOMATION FOR POWER SITES WITH THE VOLTAGE UP TO 750 KV

RPA



RPA microprocessor cabinets are designed for use as an integrated system of relay protection and automation of station and substation equipment, generating installations of various connections 6(10)-500 kV.

Cabinets are made in the form of one or two identical semi-sets, both according to the standard and according to the individual project of the Customer for non-standard solutions.

as well as:

RPA cabinets for power sites 35-500kV

- Differential protection of main busbar;
- Differential protection of the busbar;
- Differential-phase protection of 110-750 kV lines;
- Differential protection of 110-220 kV lines with FOCL;
- Directional high-frequency protection of 110-330 kV lines;
- Remote and current directional protection of 35-330kV lines, including RF blocking;
- Protection of the sectional and busbar 35-330 kV switch;
- Protection of three and two-winding transformer, autotransformer 35-500 kV;
- Protection of booster and control transformers;
- Protection of current-limiting reactors;
- Protection of the generator-transformer unit;
- Protection of the generator;
- Protection of the line to the static capacitors battery;
- Organization of circuits and protection of the voltage transformer;
- Automation of fire extinguishing of autotransformer;
- Automation of voltage regulation of power transformer and autotransformer;
- Automatic frequency unloading;
- Automatic switch control 35-220 kV;
- Fault location device
- Protection and automation panels on metal structures

General purpose cabinets:

- Central alarm cabinet;
- AC auxiliary cabinet;
- Control cabinet for switching devices with mnemonic diagram;
- Cabinet for arrangement of voltage transformers secondary circuits;
- Electricity metering cabinet;
- Cabinet for arrangement of electromagnetic blocking circuits;
- Cabinet for arrangement and distribution of direct operating current circuits.

Outdoor cabinets:

- RZA cabinets of ShNU type
- Cabinets of SHOV, ShZV, ShPPR, ShPPV, ShZST, ShZTT, ShZTN series
- Disconnectors control cabinets of SUR series

CABINETS AND EQUIPMENT FOR COMMUNICATION SYSTEMS



Communication system cabinets are designed to organize communication channels for AIM system, ACS (telemechanics), RPA, ECA systems, as well as to organize technological and operational dispatch communications at various enterprises in the electric power industry. Cabinets are manufactured in height 18U÷45U and width (depth) 600mm÷1000mm. The line of communication systems cabinets includes the following types of cabinets:

- UKSN-T-A. High-frequency communication cabinet.
- UKSN-T-V. Fiber optic transmission cabinet.
- UKSN-T-S. Industrial PBX.
- UKSN-T-D. Radio-relay cabinet.
- UKSN-T-E. Cabinet of other telecommunication systems.

SOLUTIONS FOR TELEMECHANICS, SCADA, ACS, VIDEO WALL

The video system is designed to display the parameters of tele-signaling and telemetry, and is also an auxiliary tool for the perception of the operator's control and monitoring actions.

The system can be used in control rooms of distribution networks, power plants and substations, industrial facilities, power systems and electric transport.

Due to the high resolution, the screens provide more information, such as complete diagrams of thermal, electrical and transport networks, providing multiple viewpoints.



LV switchgears

NKRU series

Low-voltage switchgears NKRU series are used both individually and as switchgear in all areas of energy consumption that require input and distribution of electric energy. NKRU are manufactured and supplied both in single-cabinet and multi-cabinet versions.

Key specifications

Rated current of main busbars

400 – 6300 A

Rated current of distribution busbars

400 – 3200 A

Rated AC voltage of main circuit

220, 380, 440, 480, 660, 240, 415, 460, 500, 690 B

Arrangement:

unilateral and bilateral maintenance

Sectioning

up to 4b



LV switchboards

SchO-2015 series

Switchboard cabinets SCHO-2015, with distribution panels, are designed to complete switchboards for the distribution of electrical energy of three-phase AC 380 V, 50 Hz. They are used to receive, distribute electrical energy, to protect against overloads and short circuit currents.

Key specifications

Rated voltage

0,4 kV

Rated current

up to 4000A



Sf6 circuit breaker

LWU36-110 series

Outdoor gas-insulated column circuit breaker is designed to turn on and off the power under load, and to protect the power system.



Key specifications

Rated voltage	Rated current	Rated withstand current
110 kV	up to 3150 A	40 kA, 4 s

As well as 220 kV SF6 circuit breakers with rated current up to 4000 A, It 50, 63 kA

Vacuum circuit breaker

ZWU-35 series

Outdoor vacuum circuit breaker circuit breaker is designed to turn on and off the power under load, and to protect the power system



Key specifications

Rated voltage	Rated current	Rated withstand current
35 kV	up to 2500 A	25; 31.5 kA, 3 s

Disconnecter

RGP-220

series

Outdoor horizontal 220 kV disconnectors, RGP-220 series are designed to turn on and off de-energized sections of high voltage electrical circuit. They are also used to cut off no-load currents of transformers and charging currents of overhead and cable lines.

as well as:

[RGP 110 series](#)

[RGP 35 series](#)

[RDZ 35 series](#)

[RLND 10 series](#)

Key specifications

[Rated voltage](#)

220 kV

[Rated current](#)

up to 4000 A

[Rated withstand current](#)

31,5; 40; 50; 63 kA



VACUUM CIRCUIT BREAKER

NV-12 series



NV-12 series indoor vacuum circuit breaker with spring-actuated drive is designed to turn on and off the power under load and protect the 10 kV power system. It can be completed as part of switchgear, KSO, reclosers, as well as for retrofit.

Key specifications

Rated voltage

10 kV

Rated current

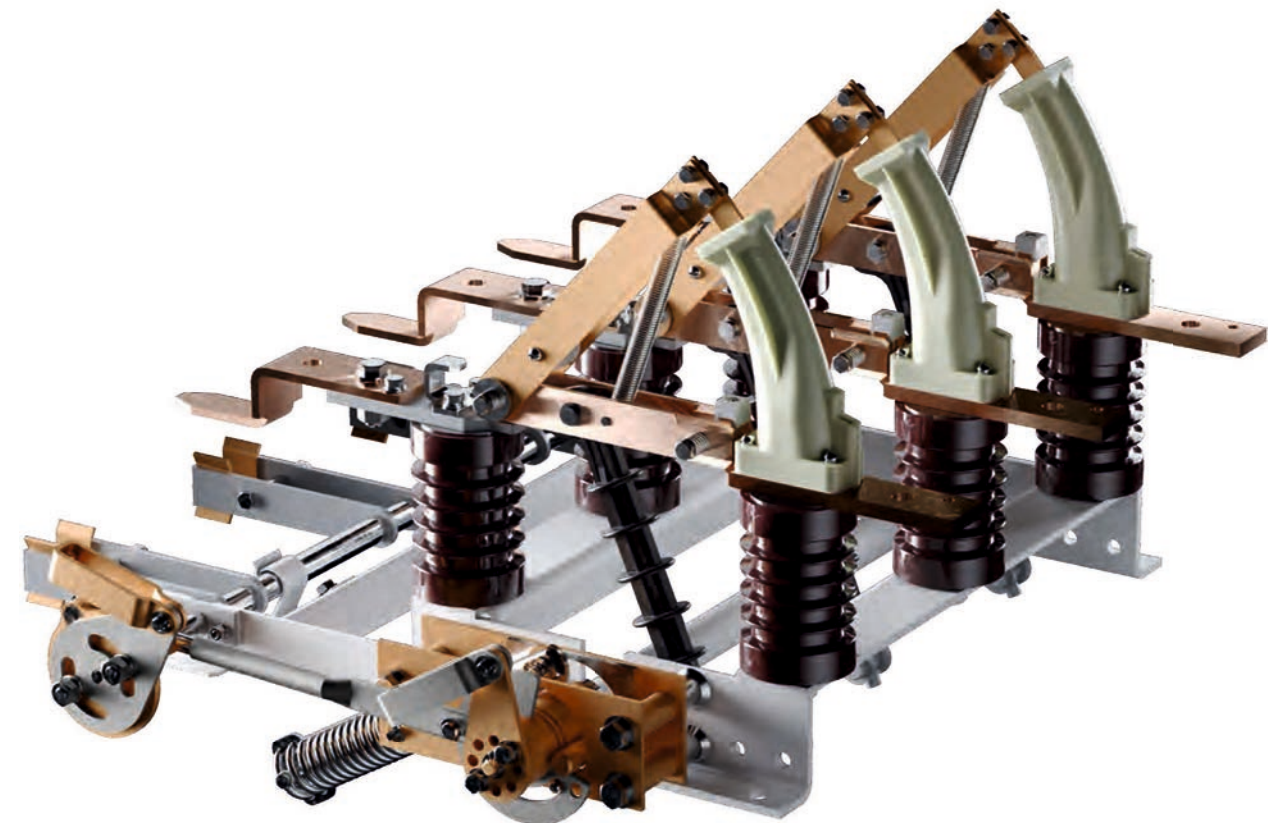
up to 4000 A

Rated breaking current

up to 40 kA

10 kV AUTOGAS LOAD BREAKER

series VNA



10 kV autogas load breakers, VNA series, are designed for switching under load of three-phase current circuits 630 A, 50 Hz. They are used in switchgear cabinets, complete transformer substations of indoor installation and prefabricated chambers of one-sided service.

Key specifications

Rated voltage

10 kV

Rated initial value of the periodic component

20kA

MODULAR BUILDINGS

Modular buildings for electrical equipment - ready-made solutions for the placement of various equipment. The experience of the enterprise allows us to design and produce modular buildings based on sandwich panels

Building Options

Modules for placing relay protection and automation cabinets, emergency automation, communication cabinets, auxiliary cabinets, control current and SCADA.

ZRU modules with switchgears up to 35 kV.

Modules for transformer substations up to 35 kV.

Modules with capacitor units.

Modules for inverter and converter blocks.



LED lighting



SKU 50-04U1

Power 50 W	Control system RF radio controller
Voltage 100 - 270 V	Size 355 × 254 × 108 mm
Frequency 50 Hz	Installation console size from Ø40 to Ø57 mm
Luminous flux 5 250 lm	Weight 1.8 kg
Luminous efficiency 105 lm/W	Operating temperature range -30°C to +60°C
Color temperature 3000 - 4000 K	Degree of protection IP 66
Color rendering index 80 R _a	



SKU 100-04U1

Power 80/100 W	Control system RF radio controller
Voltage 100 - 270 V	Size 545 × 430 × 180 mm
Frequency 50 Hz	Installation console size from Ø40 to Ø57 mm
Luminous flux 10000/12500 lm	Weight 6.5 kg
Luminous efficiency 125 lm/W	Operating temperature range -30°C to +60°C
Color temperature 3000 - 4000 K	Degree of protection IP 66
Color rendering index 80 R _a	



SKU 130/150/200/240-04U1

Power 130/150/200/240 W	Control system RF radio controller
Voltage 100 - 270 V	Size 665 × 480 × 180 mm
Frequency 50 Hz	Installation console size from Ø40 to Ø57 mm
Luminous flux 16250/28800 lm	Weight 8 kg
Luminous efficiency 130-120 lm/W	Operating temperature range -30°C to +60°C
Color temperature 3000 - 4000 K	Degree of protection IP 66
Color rendering index 80 R _a	



LV control cabinet for outdoor lightning IPNO-DU [250A]

Rated voltage AC 380 V
Rated cabinet current up to 250 A
Frequency 50 Hz
Rated current of introductory fuses up to 250 A
Rated current of outgoing fuses up to 100 A
Rated current of current transformers up to 150/5 A
Number of outgoing lines 6
Overall dimensions, mm, not more than (H×W×D) 1900 × 900 × 550 mm
Weight not more than 200 kg
Weight with AVR no more than 250 kg

Outdoor lighting devices SKU-50 and SKU-100 are designed to illuminate roads and streets with low and medium traffic intensity. For roads and streets with high traffic intensity, SKU-150 is used. Both types of devices are suitable for lighting railway platforms, parking lots, as well as for areas adjacent to public and residential buildings.

IPNO-DU* is a cabinet for remote control of outdoor lighting, designed to distribute phase voltages from a transformer substation, power and protect the street lighting network from overloads and short circuits, as well as accounting for consumption and preventing theft of electrical energy.

*IPNO-DU with AVR

SBO series street lamps



Power
60, 80, 100, 120, 150 W

Luminous flux, lm
9715, 12500, 15130, 20560, 25010

Voltage
100-277 V



TE-71 SERIES SINGLE-PHASE MULTIFUNCTIONAL ELECTRICITY METERS

Rated voltage 230 V ± 20%	Power consumed by voltage circuits, no more than: 0.1 W	Average service life of the meter, not less than: 30 years
Mains frequency: 50Hz ± 2%	Power consumed by current circuits, no more than: 0.2 VA	Dimensions: 208×130×79mm 209×130×79mm
Rated current 5; 10 A	Operating temperature range: -40 to +70°C	Weight not more than 0.9 kg
Maximum current 60; 80 A	Main communication interface: RS-485; optical port (PLC, GSM, RF)	
Accuracy class active energy 1	Sensors: housing opening / magnetic field / terminal block cover	
Accuracy class reactive energy 1	Degree of protection: IP 54	
Starting current 0.004 I _n	Battery life, not less: 10 years	

Electricity meters (TE 71; TE 73 series) are designed to measure active and reactive electricity, evaluate active and reactive power in single-phase and three-phase AC circuits of industrial frequency, automated metering, collection and remote transmission of accounting data to a single information center with the purpose of controlling electricity consumption and preparing data for conducting commercial settlements for consumed electricity on the basis of multi-tariff accounting, differentiated by time of day.



TE-73 THREE-PHASE MULTIFUNCTIONAL ELECTRICITY METERS

Rated voltage: for transformer-connected meters 3 × 57.7/100; 3×127/220; 3 × 230/400; (3 × 57.7/100...230/400V)	Accuracy class of active energy: for transformer-connected meters 0.2S; 0.5S;	Operating temperature range -40 to + 70 °C
Rated voltage: for direct connection meters 3 × 230/400 V	Active energy accuracy class: for direct connection meters 1	Sensors: housing opening / magnetic field / terminal block cover
Working voltage range: 0.7 Unom-1.2 Unom	Reactive energy accuracy class: for transformer-connected meters 0.5; 1;	Degree of protection IP 54
Rated current: for meters of transformer connection 1; 5 A	Reactive energy accuracy class: for direct connection meters 1	Battery life at least 10 years
Rated current: for direct connection meters 10 A	Pulse input current no more than 2 kA	Average service life of the meter, not less than 30 years
Maximum current: for meters of transformer inclusion 6; 10 A	Impulse voltage no more than 4 kV	Dimensions: 285×170×90mm 290×175×89.5mm
Maximum current: for direct connection meters 100 A	Communication interfaces: RS 485; optical port (PLC, GSM, RF)	Weight no more than 3 kg

DATA CONCENTRATOR

TEC70 SERIES



Rated voltage
3 x 230/400 V

Working voltage range
80% - 120%

Communication interfaces:
PLC, GPRS, RS-485, Ethernet, optical port, RS-232, USB port

Maximum number of metering devices
2000

INSTRUMENT TRANSFORMER

TT-0,66 SERIES



Rated voltage
up to 690V

Rated primary current
from 20 to 3000A

Accuracy class of secondary windings
0.5s, 0.5, 10P

EV CHARGER

GreenTE series



AC stations
single-phase up to 7kW
three-phase up to 23kW

DC stations
up to 120kW



PHOTOVOLTAIC STATION

A photovoltaic station is a system that converts the energy of sunlight into electrical energy and accumulates it in batteries.

There are two main types of photovoltaic stations: networked or connected to an external power grid (on-grid) and autonomous or independent of an external power grid (off-grid).

Solutions:

For substation auxiliaries

20 – 100 kW

For household objects

up to 1 MW

For generation to the city network

up to 500 MW



voltage 6; 10; 20; 35 kV

TMG series

THREE-PHASE DISTRIBUTION TRANSFORMERS

Power transformers, three-phase, two-winding sealed with natural oil cooling. Transformers are designed to convert electricity in networks of electricity consumers. The windings of the active part of the transformer are made in aluminum and copper versions.



Power range
25 ÷ 2 500 kVA

HV winding rated voltage
6; 10; 20; 35 kV

LV winding rated voltage
0.4 kV

Windings connection group
25 ÷ 250 kVA - Y/Yn-0; Δ/Yn-11; Y/Zn-11
400 ÷ 2500 kVA - Y/Yn-0; Δ/Yn-11

voltage 35 kV

TDNS series

TWO-WINDING POWER TRANSFORMERS WITH OLTC

Power transformer, three-phase, two-winding with natural circulation of oil and forced air, with voltage regulation under load (OLTC), with a regulation range of $\pm 8 \times 1.5\%$ from the HV side. Automatic control is carried out from an automatic controller supplied with the transformer. Transformers are intended for work in auxiliary electric networks of power plants.

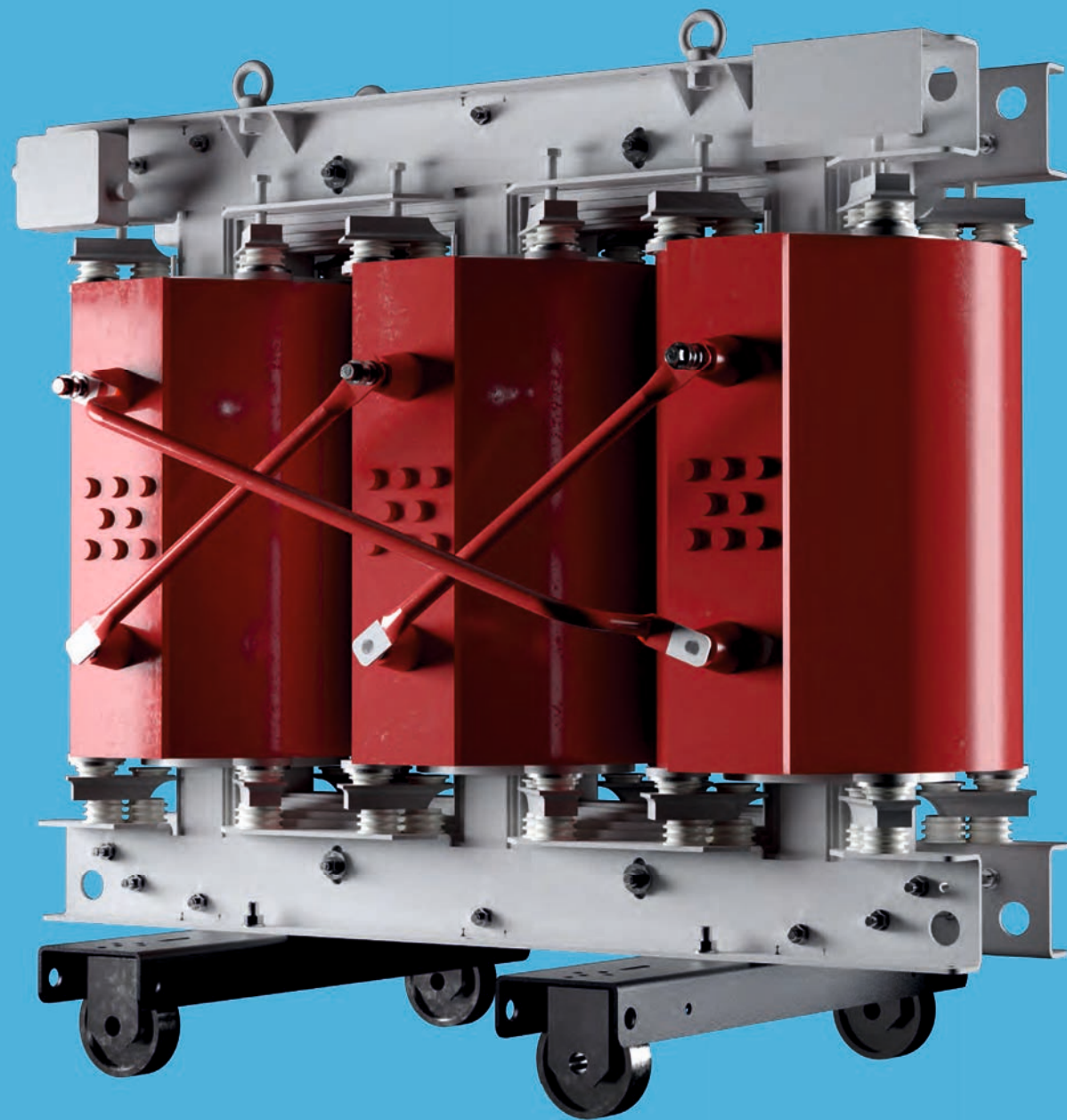


Power range
10000 ÷ 16000 kVA

Primary winding voltage HV
36.75 kV

Secondary winding voltage LV
6.3; 10.5 kV

Windings connection group
Y/Δ-11



voltage 6; 10; 20; 35 kV

TSL series

DRY TYPE DISTRIBUTION POWER TRANSFORMERS WITH CAST RESIN

TSL series transformers with power range 63 - 2500 kVA, two-winding with natural air cooling and cast resin with non-excited tap changer. They are made for areas with a temperate climate with ambient temperatures from -45 to +50°C.

Designed for operation in general purpose installations, for transformer substations and in converting installations.

Power range

63÷2 500 kVA

Windings connection group

$\Delta/Yn-11$; $Y/Yn-0$

HV winding rated voltage

6; 10; 10.5; 20; 35 kV

LV winding rated voltage

0.23; 0.4; 0.69 kV



Voltage 6; 10; 10,5; 20; 35 kV

TSZL, TSZP series

**DRY TYPE DISTRIBUTION
POWER TRANSFORMERS
WITH CAST RESIN**

Dry non-combustible, distribution power transformers with cast resin with a power of 63 - 2500 kVA, heat resistance class - "H", two-winding with natural or forced air cooling, protected design, with switching taps without excitation.

It is made for areas with a temperate climate with an ambient temperature from -45 to +50°C.

The transformers have a flat-laminated magnetic system assembled using the Step-Lap technology from high-quality electrical steel.

The active part of the transformer is protected by a casing, which is a prefabricated metal structure.

Transformers TSZP - 1600/10.5 are designed for the needs of the underground metro.

Key specifications for TSZL

- Power range
63÷2 500 kVA
- HV winding rated voltage
6; 10; 10.5; 20; 35 kV
- LV winding rated voltage
0.23; 0.4; 0.69 kV
- Windings connection group
Δ/Yn-11; Y/Yn-0

Key specifications for TSZP

- Power range
1600 kVA
- HV winding rated voltage
10.5 kV
- LV winding rated voltage
0.67; 0.67-0.67 kV
- Windings connection group
Y/Δ-11; Y/Δ-Y-11-0

voltage 35 kV

TMN series

**TWO-WINDING POWER TRANSFORMERS
WITH OLTC**

Two-winding three-phase power transformers, with natural oil circulation, with on-load tap changer. Automatic control is carried out from the automatic controller supplied with the transformer.



Power range

1000÷6300 kVA

HV winding rated voltage

35 kV

LV winding rated voltage

6,3; 11 kV

Windings connection group

Y/Δ-11

voltage 110 kV

TDTN series

**TWO-WINDING POWER TRANSFORMERS
WITH OLTC**



Power range

10000÷63000 kVA

HV winding rated voltage

115 kV

LV winding rated voltage

6.6; 11 kV

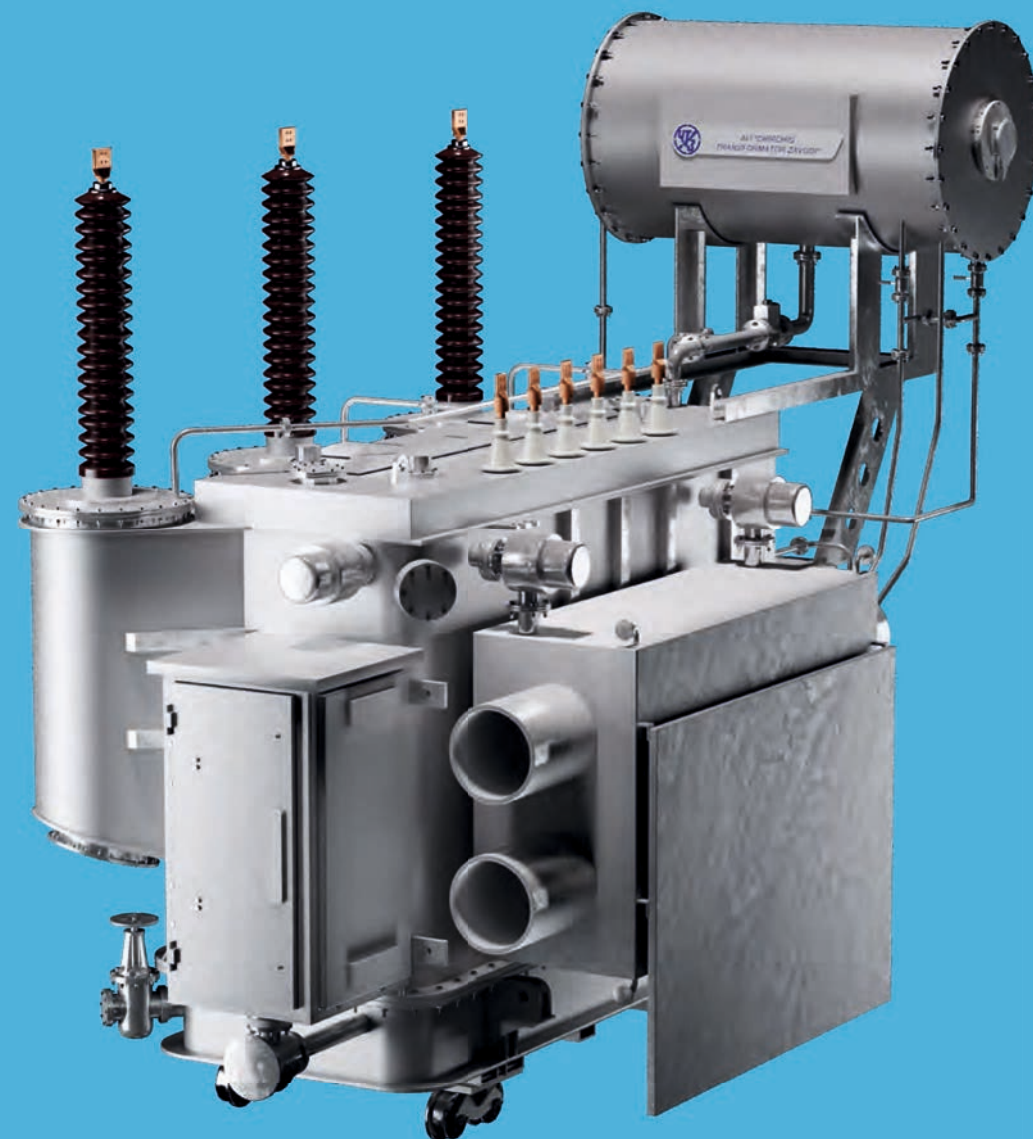
Windings connection group

Yn/Yn/Δ-0-11;

voltage 220 kV

TRDCN series

Power transformers



Power range
63000 kVA

HV winding rated voltage
220 kV

LV winding rated voltage
6,6; 11 kV

Windings connection group
Yn/Δ-Δ-11-11;

voltage 110; 220 kV

TDTNZh series

Power traction transformers



Power range
16000÷40000 kVA

HV winding rated voltage
110; 220 kV

Windings connection group
Yn/Yn/Δ-0-11;

LV winding rated voltage
6,6; 11 kV

MV windings rated voltage
27.5 kV



voltage 25 kV

SINGLE PHASE TRACTION TRANSFORMERS

ONDCE series

Traction transformer ONDCE-4350/25P-U2 is designed for freight electric locomotives ES5K, 2ES5K, 3ES5K, 4ES5K, operated on railways, electrified by alternating current, voltage 25 kV, 50 Hz.

The transformer converts the voltage of the contact network into the supply voltage of the traction motors, circuits of the excitation windings of the traction motors and auxiliary circuits of the electric locomotive. In addition, the transformer ensures the implementation of the inverting mode (converting the voltage of traction motors into voltages of the contact network, excitation of traction motors and auxiliaries).

Rated power

4350 kVA

Rated voltage of the mains winding

25 kV

Windings connection group

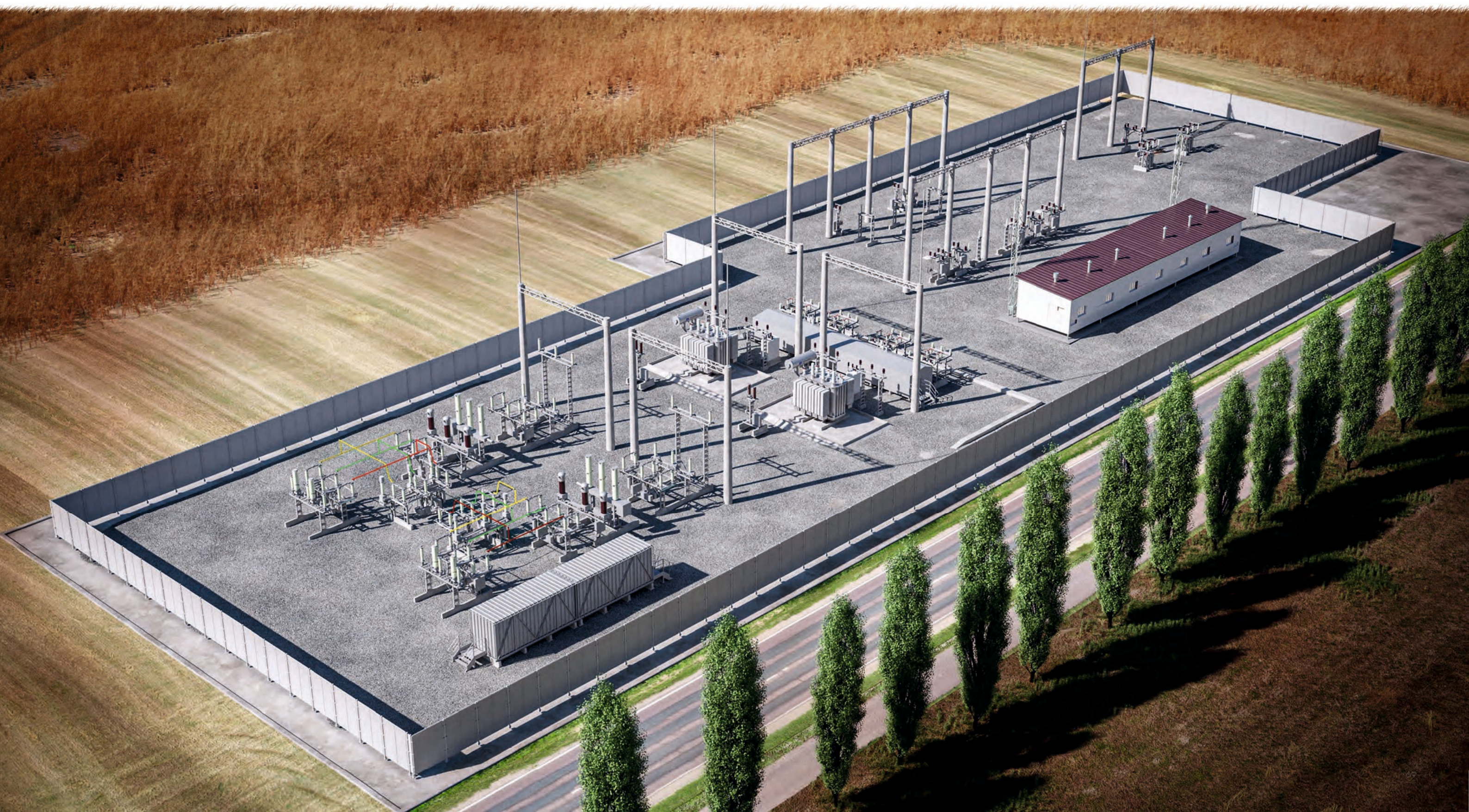
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MODULAR TRACTION TRANSFORMER SUBSTATION

WITH THE VOLTAGE 220/27.5/10 AND 110/27.5/10 kV

Traction transformer substations 20 - 220 kV for electric grid and traction power supply are designed to receive, convert and distribute electric energy of three-phase alternating current with a frequency of 50 Hz at rated voltages of 220, 110, 35, 27.5, 20 and 10 (6) kV.

They can be used both on the side of network substations and at power stations under related conditions.



BZP-13

BZP-13 is a digital relay protection device that provides protection and automation, as well as control, diagnostics and signaling at power facilities with a voltage of 0.4 - 220 kV.

The device is intended for installation in high-voltage KSO, KRU, KRUN cubicles, in relay protection cabinets, on relay panels and control panels of power stations and power grid substations, utilities and industrial enterprises, at oil and gas facilities, railways and subways.



BZP-02

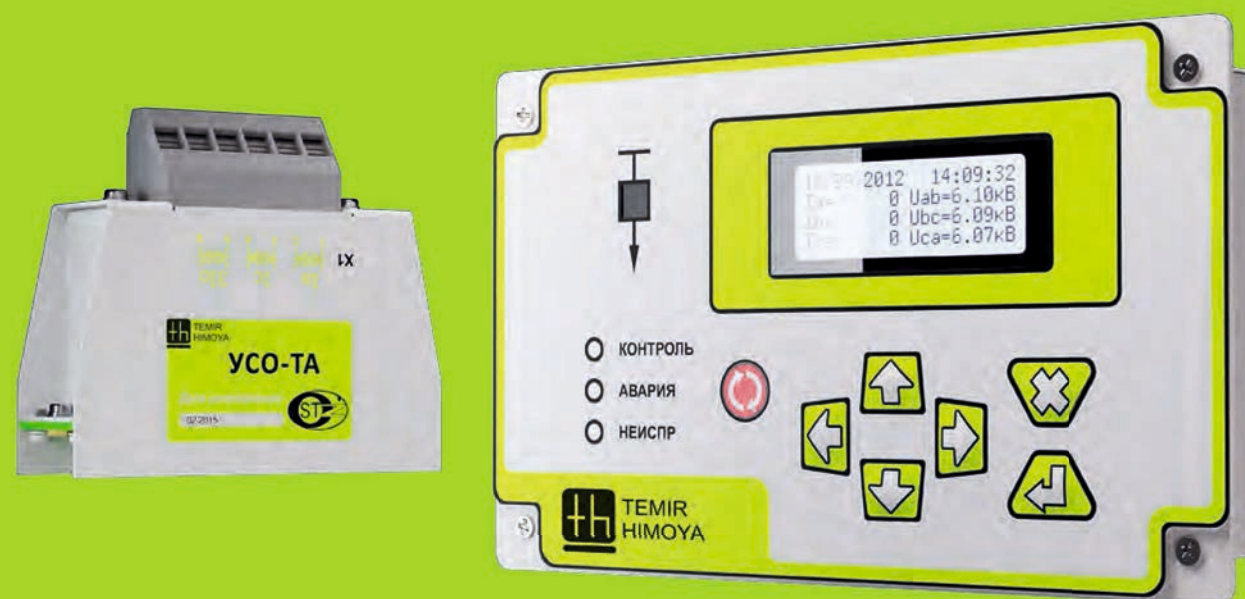
BZP-02 is a simple and functional set of relay protection and automation with current and voltage inputs, consisting of a protection unit that performs the main functions of the device. It provides connection of the USO-TA module to current circuits, arc protection sensors and control panel for configuring and viewing device parameters.

This integrated solution makes it possible to perform the functions of ACR automation and selective arc protection of RU-6(10) kV without additional relay protection devices for general purpose. BZP-02 has established itself as a simple and reliable solution for protecting substation equipment.



BZP-01

BZP-01 is designed to perform the functions of relay protection, automation, control, diagnostics and signaling of connections with a voltage of 6-35 kV at facilities with alternating or direct operating current, rated voltage of 110 or 220 V. The device can be used for connections of overhead and cable lines transmission lines, transformers and electric motors. The device is intended for installation in high-voltage cubicle of KSO, KRU, KRUN, KTP, etc., on relay panels and control panels of power plants and substations of electric grid, municipal and industrial enterprises. A feature of BZP-01 is its versatility, one unit without changing the firmware can be installed on all types of feeder connections: KL, VV, SV, TN.



MI-96

Digital measuring device MI-96 is designed to measure and indicate the values of electrical quantities of the operating modes of three-phase AC electrical networks with a nominal frequency of 50 and 60 Hz.

The device supports configuration via Android phone via Bluetooth 4.2 interface, oscillogram recording, monitors and measures 30 electrical quantities.

It has discrete input/output, RS-485 digital interface and two rows of seven-segment indicators.



BZP-ZDZ-01

BZP-ZDZ-01 is designed to protect KRU, KRUN and KSO switchgears of 0.4–35 kV power stations and substations from short circuits accompanied by an open electric arc.

To implement the protection of one cubicle of 6-35 kV switchgear, it is necessary to install BZP-ZDZ-01 with three fiber-optic sensors, each of which is located in an isolated compartment.

The scope of BZP-ZDZ-01 is power stations and substations, power supply facilities for the gas and oil industries, industrial enterprises, subways and traction substations of electrified railways.

The device is easily integrated into the protection system of the energy facility and is designed to work together with all microprocessor relay protection terminals on the market, as well as with protection made on electromechanical relays.

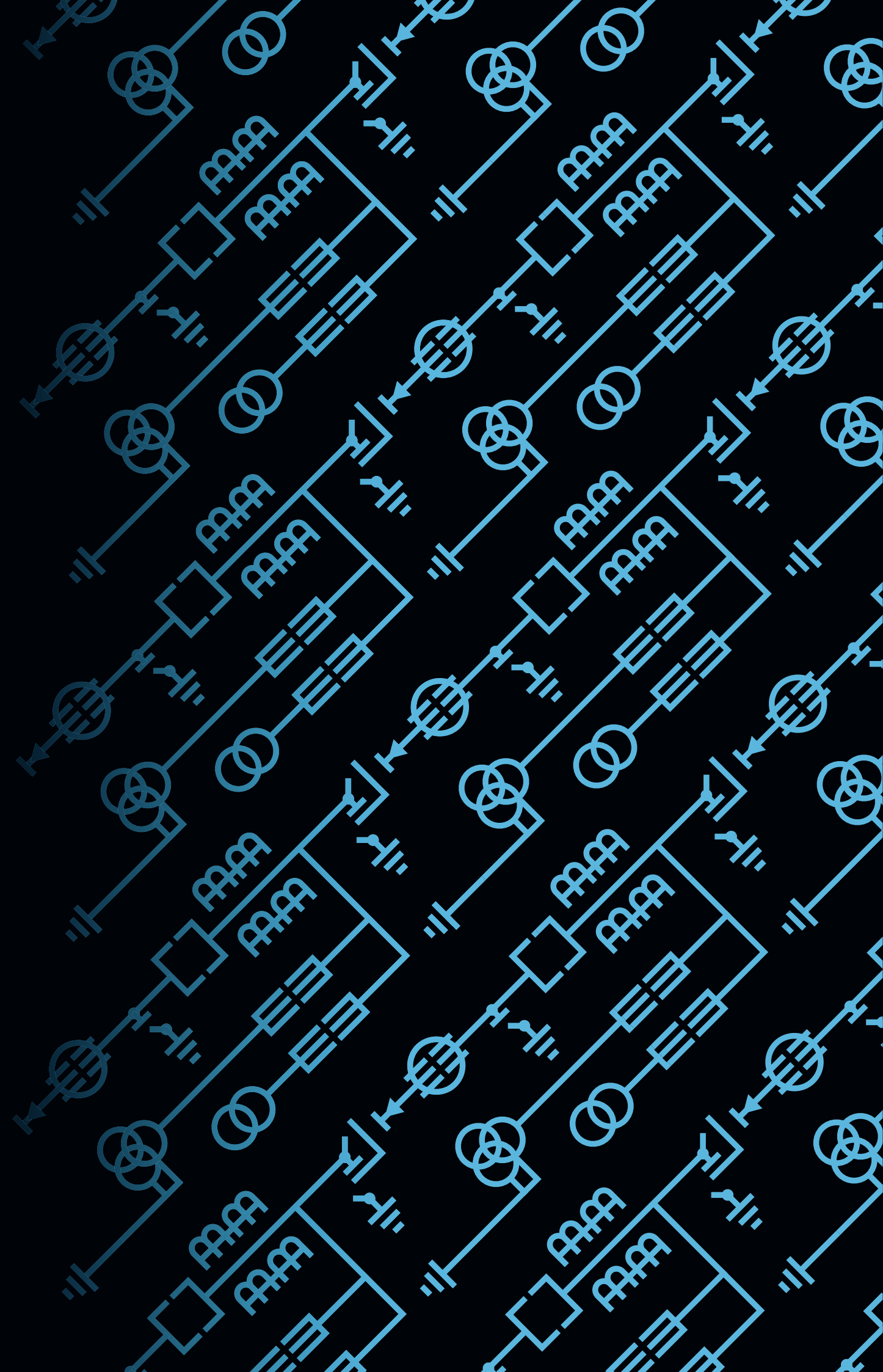


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